Cardiovascular stent design and wall shear stress distribution in coronary stented arteries

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The stent is a major breakthrough in the treatment of coronary artery diseases. The permanent vascular implant of a stent, however, changes the intra-stent blood flow haemodynamics. There is a growing consensus that the stent implant may change the artery wall shear stress distribution and hence trigger the restenosis process. Computational fluid dynamics (CFD) has been widely used to analyse haemodynamics in stented arteries. In this Letter, CFD models were developed to investigate the effects of stent design pattern and strut geometry, respectively, on the wall shear stress distribution in coronary stented arteries. Assessment of the potential restenosis risk was primarily based on the wall shear stress distribution. Results show that the stent design pattern alone does not have a significant impact on the stent haemodynamic behaviour. Wall shear stress is very sensitive to strut thickness, while varying the strut width or crown radius has very little effect. The proposed methodology and findings will provide great insight for future optimisation of stent design to reduce the risk of restenosis.

1. Introduction: Coronary stenting has emerged as the primary treatment for cardiovascular diseases and has received great attention from the medical community since its introduction in the 1990s. Over the last two decades, the field has seen numerous innovations in attempts to perfect the percutaneous management of coronary artery diseases. The introduction of drug eluting stents in the last decade has resulted in dramatic lowering of restenosis (re-narrowing of the arteries) after angioplasty, leading to worldwide acceptance of this new invention for coronary artery diseases. With this revolutionary technology, stents today have extended their indications from coronary artery diseases to many other indications, including carotid, renal, iliac and femoral arteries.

A stent is a small, coiled wire-mesh tube that can be deployed in an artery and expanded percutaneously using a balloon catheter during angioplasty to open a narrowed or blocked artery to restore blood flow from atherosclerotic vascular diseases. However, after stenting, restenosis may occur if the newly developed neo-intimal tissues on the inner lining of the arteries proliferate excessively. Although restenosis is a complicated physiological phenomenon, several studies have shown that the stent design could be one of the critical factors affecting restenosis [1–3]. A growing number of experts suspect that restenosis may be a result of altered haemodynamics or, more precisely, the changes in wall shear stress distribution in arteries after stenting. Given the role of wall shear stress in the development of atherosclerotic plaques [4, 5], it is reasonable to assume that it may also contribute to the neo-intimal hyperplasia formation. Furthermore, clinical studies have reported that low or oscillating wall shear stress has been observed at the location where neo-intimal thickening in stented arteries is the greatest [6–10].

In recent years, computational modelling has emerged as an important tool for the optimisation of stent design and can be used along with experimental data to improve the stent performance. Such models could give insights into various aspects of stent design that may consequently reduce the risk of vascular injury and subsequent restenosis. Today, many researchers use computational fluid dynamics (CFD) to predict the wall shear stress distribution in stented arteries and the potential risk of restenosis. In these studies, wall shear stress acting on the artery wall is analysed using certain critical threshold values produced by several *in vitro* and *in vivo* studies. These Letters have suggested a strong inverse correlation between low wall shear stress (less than 5 dynes/cm^2) and atherosclerotic neo-intimal thickening [11, 12]. However, elevated shear stress (more than 15 dyne/cm^2) induces a more quiescent and anti-proliferative phenotype [4, 5].

Even though wall shear stress plays such a key role in the development of restenosis, its relationship with the stent design parameters (e.g. design pattern and strut geometry) is yet to be systematically investigated. To date, it is uncertain as to whether specific stent design changes would result in significant wall shear stress variations or not. Therefore in this study, three-dimensional CFD models were developed to investigate the effects of stent design on the wall shear stress distribution in stented arteries and to determine which design parameter is the most dominant one. The proposed methodology can be used to evaluate the haemodynamic performance of a stent in attempts to reduce its restenosis risk.

2. Methodology

2.1. Governing equations: The flow was governed by flow equations for conservation of mass and momentum, given in (1) and (2), respectively:

$$\nabla \cdot v = 0 \tag{1}$$

$$\rho\left(\frac{\partial v}{\partial t} + v \cdot \nabla v\right) = -\nabla p + \mu \nabla^2 v + f \tag{2}$$

In this study, the blood flow in a stented artery was assumed to be steady, incompressible and laminar. The governing equations were solved by the commercial CFD solver ANSYS FLUENT using its pressure–velocity coupled algorithm. This algorithm is able to solve the velocity and pressure fields simultaneously to provide robust solutions. A fully developed Hagen–Poiseuille flow of 120 ml/min [13] was prescribed as the inlet boundary condition, and uniform pressure was prescribed as the outlet boundary condition. The artery wall was assumed to be a fixed no-slip surface, and its diameter was 3 mm. Blood was modelled as an incompressible fluid with a density of $\rho = 1060$ kg/m^3. Approximately, 13 000 000 unstructured mesh grids were generated for each stented artery. All results in this study were tested to be mesh independent. The non-Newtonian characteristic of the blood was described as follows using the Carreau model

$$\mu = \mu_\infty + (\mu_0 - \mu_\infty)[1 + (\lambda \dot{\gamma})^2]^{(n-1)/2} \tag{3}$$

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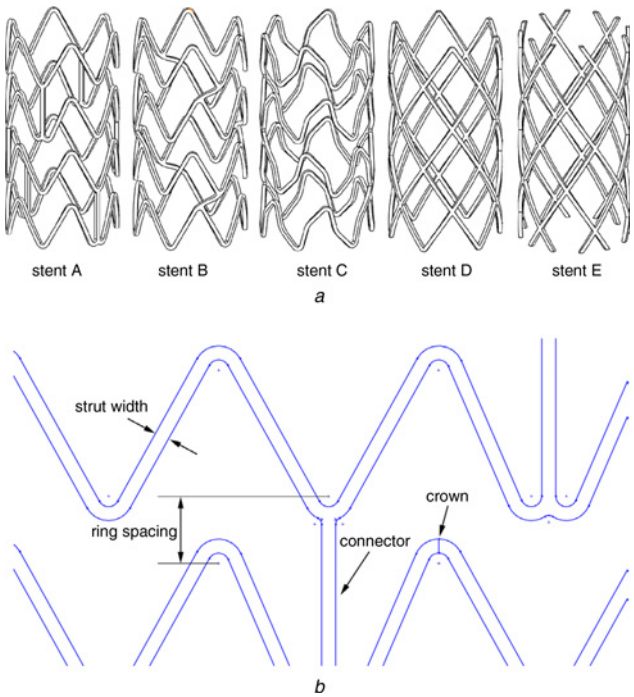

Figure 1 *Stent CFD models*
a Stents A–E for design pattern study
b Stent strut geometric parameters

where μ_∞ and μ_0 are viscosities when shear rate becomes infinity and zero, respectively, $\dot{\gamma}$ is shear rate, and λ and n are material coefficients. These parameters can be obtained by fitting the Carreau model to the experimental data [14, 15] where the fitted values are $\mu_\infty = 0.0035$ kg/ms, $\mu_0 = 0.25$ kg/ms, $\lambda = 25.00$ s and $n = 0.25$.

As aforementioned, the wall shear stress is strongly related to restenosis and will be adopted as the major criterion in this analysis. The magnitude of the wall shear stress vector is calculated as

$$\tau_w = \mu\left[2\left(\frac{\partial u}{\partial x}\right)^2 + 2\left(\frac{\partial v}{\partial y}\right)^2 + 2\left(\frac{\partial w}{\partial z}\right)^2 + \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}\right)^2\right]^{(1/2)} \quad (4)$$

where u, v and w are the x, y and z components of velocity vector, respectively.

2.2. Stent CFD models: This analysis was divided into two major parts, with each part seeking to investigate the effects of stent design pattern and strut geometry, respectively, on the wall shear stress distribution in coronary stented arteries. To assess the effects of stent design pattern alone, four types of stents, which resembled the four most widely used commercial coronary stents on the market, were studied in this Letter. Fig. 1*a* shows the design patterns of these four stents, referred to as stents A, B, C and D. Unlike stents A–C, the design patterns of which are typically laser-cut directly onto the micro-sized hypotube, stent D is a braided stent in a tubular mesh configuration composed of monofilament wires. Such a braided configuration could sometimes leave untrimmed edges at the stent ends and cause undesired issues. Therefore stent E has the same configuration as stent D, but with a 25% offset axially to represent the case of untrimmed edges. To focus on the effects of stent design pattern alone, stents A–C were intentionally constructed with the same strut geometries. Their corresponding strut geometries, such as strut width, strut thickness, crown radius and ring spacing, as shown in Fig. 1*b*, were constant for all, so only the design pattern effects were isolated for evaluation. Stents D and E had the same ring number as stents A–C, except that their ring spacing and crown radius definitions were slightly different because of the constraints of their braided design, as shown in Figs. 1 and 2.

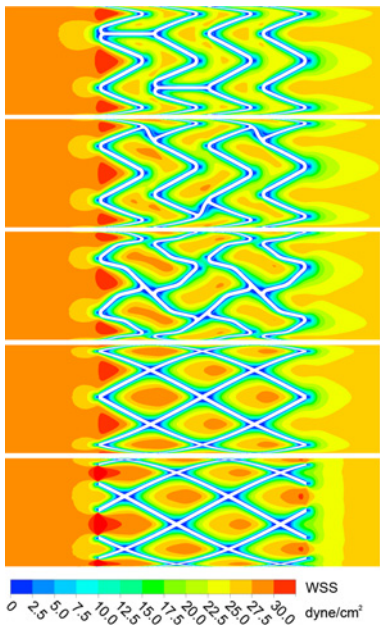

Figure 2 *Contour plots of wall shear stress distribution on artery wall after implantation of coronary stents A–E (from top to bottom)*

To assess the effects of stent strut geometry alone, a parametric analysis was conducted to systematically assess the effects of varying strut geometries on the wall shear stress distribution. Only stent A was selected for evaluation in this part of the analysis. The selected strut geometries for investigation included (1) crown radius (CR), (2) strut width (SW) and (3) strut thickness (ST), as defined in Fig. 1*b*. Each parameter was varied in its dimensions from −30 to +30% (compared to the standard case) for sensitivity study or trend analysis of the stent strut geometries and wall shear stress.

3. Results and discussion

3.1. Effects of stent design pattern: Contour plots of the wall shear stress distribution on the artery wall after implantation of coronary stents A–E (from top to bottom) are illustrated in Fig. 2. The results show that the wall shear stress contour plots were quite similar in the upstream, stented and downstream regions for all stents investigated. The colour blue indicates low wall shear stress, which was our major interest in this study. It is apparent that the minimum wall shear stress occurred at the recirculation zones located at the downstream or backside of each stent strut or joint (connector) of the strut. Fig. 3*a* shows the streamline plots developed around the crown/connector regions of stent B (left) and stent C (right), where large recirculation zones were typically observed. This agrees well with the thrombogenicity (blood clot test) results conducted internally, showing that platelet deposition initiated at the crown region first and accumulated fast, as shown in Fig. 3*b*. Several clinical studies have reported that low or oscillating wall shear stress has been observed at the location where neo-intimal thickening in stented arteries is the greatest. To quantify the effects to provide more distinct results, the total wall surface area exposed to low shear stress of 5 dyne/cm^2, below which restenosis is likely to occur [4, 5], was calculated and measured. At the flow rate of 120 ml/min, the total wall surface area with shear stress below 5 dyne/cm^2 was 4.04 mm^2 for stent A, 4.11 mm^2 for stent B, 4.17 mm^2 for stent C, 3.90 mm^2 for stent D and 4.13 mm^2 for

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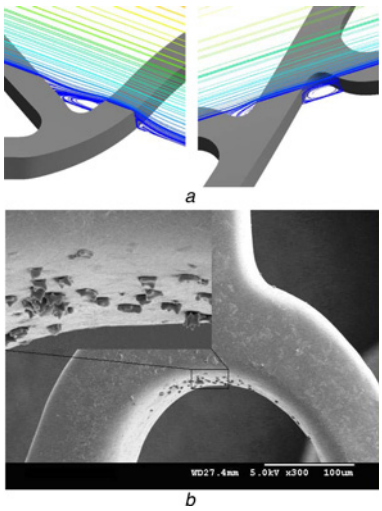

Figure 3 *Streamlines and platelet deposition*
a Streamlines around crowns/connectors of stent B (left) and stent C (right)
b Platelet deposition initiated at crown region first

Table 1 Effects of stent design pattern on low wall shear stress area

	Low wall shear stress area, mm^2
Stent A	4.04
Stent B	4.11
Stent C	4.17
Stent D	3.90
Stent E	4.13

stent E (Table 1). Based on these data, and despite their distinct stent design patterns, the difference among these five stents was very marginal, with the braided stent D showing the best haemodynamic performance.

Stents B and C had similar designs, but stent C was an asymmetric version of stent B. It had skewed struts that could deviate part of the main blood flow direction from axial to circumferential, thereby decreasing the flow velocity in the axial direction. Fig. 4 shows the velocity vector projection on the stent cross-section at the same location for stents B (top) and C (bottom). It is clear that the velocity vector projection of stent B was symmetric, whereas that of stent C was inclined more towards one side than toward the other because of its asymmetric design. The difference in the low wall shear stress area between these two stents was minor, with stent C slightly worse than stent B by 1.5%. This demonstrates that the skewed strut design does not have a significantly negative impact in terms of wall shear stress. However, it should be noted that the curved connectors in stent B seem to create more recirculation zones than the straight connectors in stent C.

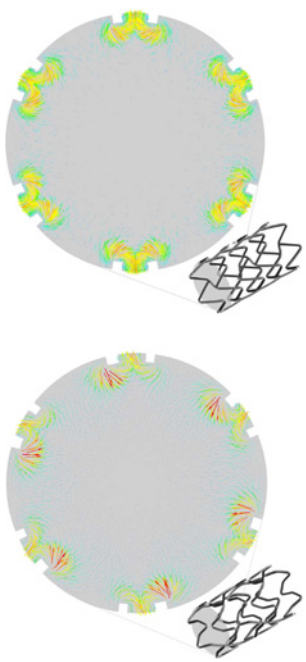

Figure 4 *Velocity vector project on same stent cross-section for stent B (top) and stent C (bottom)*

Stent E was an untrimmed-edge version of stent D, designed to represent a more realistic clinical situation. The results suggested that stent E was less ideal than stent D, with a difference of 5.9% in the low wall shear stress area. According to the wall shear stress contour plots (Fig. 2), it is reasonable to suspect that the untrimmed edges on stent E were the major reason for the difference in wall shear stress. Therefore the use of braided stents with untrimmed edges should be minimised in clinical practice.

In all, these four types of stents, resembling the four most widely used commercial coronary stents on the market, did not generate significant differences in the wall shear stress area under the condition that their corresponding strut geometries were kept the same. Therefore the stent design pattern does not have a significant impact on the distribution of wall shear stress.

3.2. Effects of stent strut geometry: Stent A was selected to systematically assess the effects of varying strut geometries alone on the wall shear stress distribution. The geometric parameter of each strut was varied in its dimension from −30 to +30% (compared to the standard case) for sensitivity studies. Fig. 5*a* and Table 2 show the simulation results of the low wall shear stress area with respect to the strut geometries, namely crown radius, strut thickness and strut width. The data show that increasing the strut thickness exposed greater areas to low wall shear stress, while varying the strut width, had very little effect. The low wall shear stress area increased slightly as the crown radius became larger. This is reasonable, as the larger crown does allow more room for the recirculation zone to fully develop.

Wall shear stress is very sensitive to strut thickness. Fig. 5*b* shows the contour plot comparison of the wall shear stress distributions for the stent with a 30% reduction (top) and 30% increase (bottom) in the strut thickness. The colour blue indicates low wall shear stresses. At the flow rate of 120 ml/min, the low wall shear stress areas were 3.25 and 5.36 mm^2 for the 30% reduction and 30% increase in the strut thickness, respectively, with a significant difference of 49% between them. This suggests that a thinner stent design could improve the stent haemodynamic behaviour, and work on such a design could become one of the future directions for stent development in the medical device world.

4. Conclusions: In this Letter, we have presented a thorough comparison of the wall shear stresses in coronary stented arteries implanted with stents A–E, which resembled the four most widely used commercial coronary stents on the market. This analysis was conducted to investigate the effects of stent design pattern and strut geometry, respectively, on the wall shear stress distribution in coronary stented arteries. The results can be summarised as follows: the stent design pattern alone does not have a significant impact on the wall shear stress distribution; the use of braided stents with untrimmed edges should be minimised in clinical practice; wall shear stress is very sensitive to strut thickness, while varying the strut width or crown radius has very little effect; a thinner stent design could improve the stent haemodynamic

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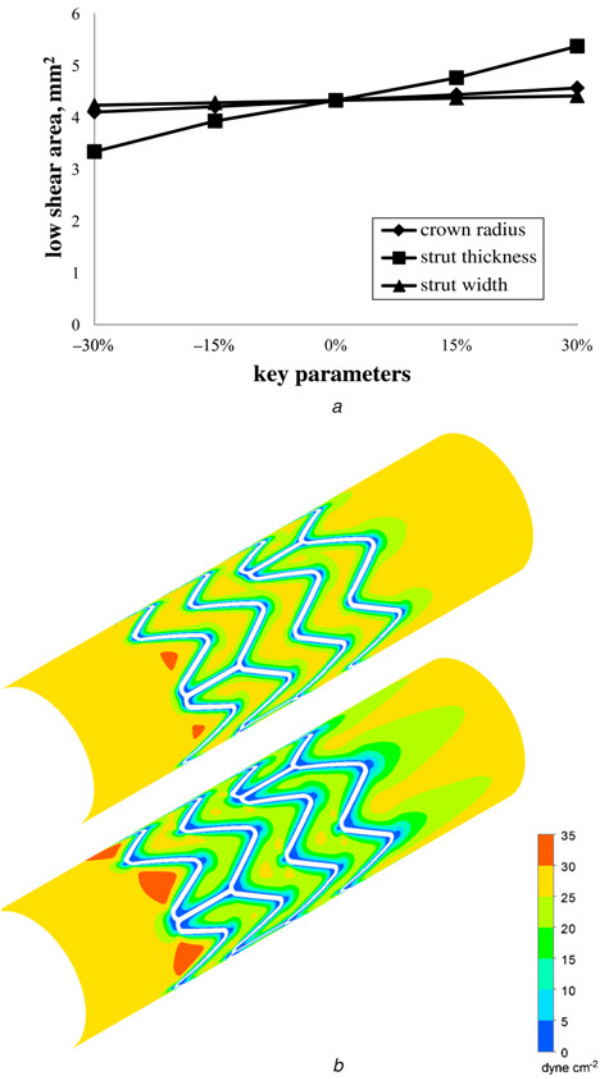

Figure 5 *Effects of stent shunt geometry on wall shear stress*
a Low wall shear stress area as a function of stent strut geometries
b Contour plots of wall shear stress for 30% reduction (top) and 30% increase (bottom) in strut thickness values

Table 2 Effects of stent strut geometries on low wall shear stress area (mm^2)

Stent design parameter	−30%	−15%	Standard	+15%	+30%
Crown	3.82	3.95	4.04	4.15	4.29
Strut thickness	3.25	3.62	4.04	4.68	5.36
Strut width	3.93	3.99	4.04	4.09	4.15

behaviour; the search for such a design could become one of the future directions for stent development in the medical device world.

These conclusions are very important for the optimisation of future stent design to help achieve the best haemodynamic behaviour.

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